S M (Rifat) Rafiuddin

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Natural Language Processing | Efficient Language Models | Interpretable AI

EDUCATION

Oklahoma State University Stillwater, OK
Ph.D. student in Computer Science *August 2022 – Expected July 2027*

- **Advisor:** Dr. Atriya Sen; **Qualifying examination:** passed September 2026.
- **Dissertation:** *Adaptive Computation in Transformers: From Selective Information Processing to Causal Mechanistic Interpretability*

Rajshahi University of Engineering and Technology Rajshahi, Bangladesh
B.Sc. in Computer Science and Engineering *January 2012 – October 2016*

ACADEMIC APPOINTMENTS

Oklahoma State University Stillwater, OK
Graduate Teaching and Research Assistant, Department of Computer Science *August 2022 – Present*

- **Graduate Teaching Assistant:** Academic-year appointments, August–May, beginning Fall 2022.
- **Graduate Research Assistant, Complex Systems Lab:** Summers 2023 and 2024.
- **Graduate Research Assistant, Reasoning and Artificial Intelligence (rAIson) Lab:** Summers 2025 and 2026.

University of Asia Pacific Dhaka, Bangladesh
Lecturer, Department of Computer Science and Engineering *October 2018 – July 2022*

Uttara University Dhaka, Bangladesh
Lecturer, Department of Computer Science and Engineering *February 2017 – October 2018*

RESEARCH AND TEACHING PROFILE

Computer scientist and Ph.D. student developing efficient, interpretable, and controllable language models. Research connects natural language processing, adaptive computation, mechanistic interpretability, counterfactual evaluation, multimodal learning, and measurement validity. The record includes 18 peer-reviewed papers and accepted manuscripts, 13 as first or sole author, and two publication-producing mentoring projects. My independent agenda links selective computation with causal tests of model mechanisms and preservation-aware adaptation. More than five years as an instructor of record covers programming, algorithms, operating systems, machine learning, pattern recognition, graphics, compilers, and web development. Oklahoma State teaching assignments add databases, security, and computing ethics, including lead TA support for online instruction. Official UAP evaluations cover 25 sections and 647 responses, with a response-weighted mean of 4.12/5 against a 4.00/5 institutional benchmark.

SELECTED CONTRIBUTIONS IN AI AND LANGUAGE MODELS

Efficient and Inspectable Language Models 2025–Present

– Developed context-conditioned rank routing, representation-guided token selection, and adaptive memory retention, reducing adapter FLOPs by 61%, producing 1.3–3.6× speedups, and preserving at least 95% of full-model performance while retaining only 30–50% of token representations.

Preservation-Aware Counterfactual Evaluation 2024–2026

– Developed C^3T and CAVE-ABSA to test aspect- and conversation-level causal hypotheses through targeted counterfactual interventions, non-target preservation, semantic consistency, and controlled evaluation.

Measurement Validity and Political Bias in LLMs [Paper] IJCNLP–AACL 2025

– Co-authored a factor analysis showing that prompt phrasing and fine-tuning can alter political-compass measurements; examined what survey-based scores establish about model behavior.

Mechanistic Interpretability and Symbolic Reasoning [Paper] ICLR 2026

– Localized a compact controller for program-like state transitions and established causal relevance by knockout: 91.6% longest-length accuracy versus 66.3% for the strongest comparison, falling to 44.8% after knockout.

PUBLICATIONS

Publications are grouped by type and listed in descending order by publication year.

Refereed Conference and Findings Papers

[1] **S M Rafiuddin** and Atriya Sen. "C^{3}T: Counterfactual Causal Reasoning for Sentiment Shifts in Social-Media Conversation Trees." Accepted to the *2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*. **Main Conference; to appear. [Paper]**

[2] **S M Rafiuddin** and Rafae Abdullah. "Context-Conditioned Masked LoRA: Dynamic Rank Routing for Compute-Efficient Parameter-Efficient Fine-Tuning." In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 26670–26689. Association for Computational Linguistics, 2026. **[Paper]**

[3] **S M Rafiuddin** and Muntaha Nujat Khan. "Emergent Discrete Controller Modules for Symbolic Planning in Transformers." In *International Conference on Learning Representations (ICLR 2026)*, Rio de Janeiro, Brazil, 2026. **[Paper] [Presentation] [Poster] [Video]**

[4] **S M Rafiuddin**. "MaskLoRA: Low-Rank Subspace–Induced Token Masking for Efficient and Faithful Language Models." In *Findings of the Association for Computational Linguistics: EACL 2026*, pp. 5678–5692. Association for Computational Linguistics, 2026. **[Paper] [Presentation] [Poster] [Video]**

[5] **S M Rafiuddin** and Muntaha Nujat Khan. "A Formal Analysis of Chain-of-Thought Prompting via Turing Reductions." In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP–AACL 2025): Short Papers*, pp. 87–98. Asian Federation of Natural Language Processing and Association for Computational Linguistics, 2025. **[Paper]**

[6] Sadia Kamal, Lalu Prasad Yadav Prakash, **S M Rafiuddin**, Mohammed Rakib, Atriya Sen, and Sagnik Ray Choudhury. "A Detailed Factor Analysis for the Political Compass Test: Navigating Ideologies of Large Language Models." In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP–AACL 2025): Short Papers*, pp. 284–303. Asian Federation of Natural Language Processing and Association for Computational Linguistics, 2025. **[Paper]**

[7] **S M Rafiuddin**. "Edu-EmotionNet: Cross-Modality Attention Alignment with Temporal Feedback Loops." In *Proceedings of the 24th International Conference on Machine Learning and Applications (ICMLA 2025)*, pp. 200–205. IEEE, 2025. **[Paper] [Presentation]**

[8] Md. Sohanuzzaman Soad, Mahady Al Hady, **S M Rafiuddin**, and Sudip Ghose. "Cross-Domain Adversarial Augmentation: Stabilizing GANs for Medical and Handwriting Data Scarcity." In *Proceedings of the 2025 IEEE 4th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON 2025)*. IEEE, 2025. **[Paper]**

[9] **S M Rafiuddin** and Muntaha Nujat Khan. "Learning What to Remember: Adaptive Probabilistic Memory Retention for Memory-Efficient Language Models." In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 3969–3981. Association for Computational Linguistics, 2025. **[Paper]**

[10] **S M Rafiuddin**, Sadia Kamal, Mohammed Rakib, Arunkumar Bagavathi, and Atriya Sen. "AdaptiSent: Context-Aware Adaptive Attention for Multimodal Aspect-Based Sentiment Analysis." In *Social Networks Analysis and Mining: Proceedings of the 17th International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2025)*, Part I, Lecture Notes in Computer Science, vol. 16322, pp. 133–141. Springer, 2025. **[Paper] [Presentation]**

[11] **S M Rafiuddin**, Mohammed Rakib, Sadia Kamal, and Arunkumar Bagavathi. "Exploiting Adaptive Contextual Masking for Aspect-Based Sentiment Analysis." In *Advances in Knowledge Discovery and Data Mining: Proceedings of the 28th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2024)*, Part VI, Lecture Notes in Computer Science, vol. 14650, pp. 147–159. Springer, 2024. **[Paper] [Presentation]**

[12] Md. Abdul Karim, **S M Rafiuddin**, Md. Jahidul Islam Razin, and Tahira Alam. "Isolated Bangla Handwritten Character Classification Using Transfer Learning." In *Proceedings of the 2nd International Conference on Computing Advancements (ICCA 2022)*, pp. 11–17. Association for Computing Machinery, 2022. **[Paper]**

[13] **S M Rafiuddin**. "High Cursive Complex Character Recognition Using GAN External Classifier." In *Proceedings of the 2nd International Conference on Computing Advancements (ICCA 2022)*, pp. 466–472. Association for Computing Machinery, 2022. **[Paper]**

[14] Md. Jahidul Islam Razin, Md. Abdul Karim, M. F. Mridha, **S M Rafiuddin**, and Tahira Alam. "A Long Short-Term Memory (LSTM) Model for Business Sentiment Analysis Based on Recurrent Neural Network." In *Sustainable Communication Networks and Application*, Lecture Notes on Data Engineering and Communications Technologies, vol. 55, pp. 1–15. Springer, 2021. **[Paper]**

[15] **S M Rafiuddin**. "Estimation of Phylogenetic Tree Using Gene Sequencing Data." In *2019 4th International Conference on Electrical Information and Communication Technology (EICT)*, pp. 1–5. IEEE, 2019. **[Paper]**

[16] **S M Rafiuddin**. "Ranking of Bangla Word Graph Using Graph-Based Ranking Algorithms." In *2017 3rd International Conference on Electrical Information and Communication Technology (EICT)*, pp. 1–5. IEEE, 2017. **[Paper]**

[17] Sadia Zaman Mishu and **S M Rafiuddin**. "Performance Analysis of Supervised Machine Learning Algorithms for Text Classification." In *2016 19th International Conference on Computer and Information Technology (ICCIT)*, pp. 409–413. IEEE, 2016. **[Paper]**

Refereed Workshop Papers

[18] **S M Rafiuddin**, Vamsi Krishna Pavuluri, and Atriya Sen. "Constraint-Aware Counterfactual Editing for Aspect-Based Sentiment Analysis." In *HHAI-KEML 2026: 2nd International Workshop on Informing Machine Learning with Knowledge Engineering*, Brussels, Belgium, 2026. **[Paper]**

RESEARCH EXPERIENCE

Reasoning and Artificial Intelligence (rAIson) Lab Oklahoma State University
Graduate Research Assistant; Advisor: Dr. Atriya Sen *Summers 2025 and 2026*

- Organized dissertation research around a Select–Interpret–Verify framework for adaptive computation across tokens, modalities, memory states, symbolic operations, and low-rank parameter directions.
- Developed published methods for representation-guided and parameter-efficient adaptation, including token masking, adaptive memory retention, low-rank subspace intervention, and context-conditioned rank routing.
- Investigated reusable computational structures for transformer reasoning and symbolic planning; current dissertation work extends this analysis through probing, feature ablation, activation patching, interchange interventions, stability tests, and strong non-SAE baselines.
- Developed CAVE-ABSA and C^{3}T for preservation-aware counterfactual intervention, emphasizing target specificity, non-target preservation, semantic consistency, and causal evaluation.
- Built reproducible PyTorch and Hugging Face pipelines with controlled ablations and efficiency profiling; ongoing collaboration aligns crop imagery and meteorological measurements for soil-moisture prediction.

Complex Systems Lab Oklahoma State University
Graduate Research Assistant; Advisor: Dr. Arunkumar Bagavathi *Summers 2023 and 2024*

- Developed adaptive contextual masking methods for aspect-based sentiment analysis to improve aspect-specific representation learning and sentiment prediction.
- Investigated multimodal aspect-based sentiment analysis using context-aware attention and text–image representation alignment.
- Conducted model comparisons, component ablations, and error analyses to evaluate the contributions of masking, attention, and multimodal fusion mechanisms.

RESEARCH SUPPORT AND AWARDS

Research and Travel Support

ICLR 2026 Financial Assistance Award, International Conference on Learning Representations 2026

– Awarded complimentary registration and $1,600 in travel support to present research at ICLR 2026 in Rio de Janeiro, Brazil.

AIRS Travel Fund, Oklahoma State University 2025

– Received institutional support for presenting and disseminating peer-reviewed research.

GPSGA Travel and Research Award, Oklahoma State University 2024–2025

– Awarded $600 in 2024 and $300 in 2025 to support conference travel and research dissemination.

Selected Competitive Honors

Honorable Mention, ICT Fest, Islamic University of Technology 2014
Honorable Mention, National Collegiate Programming Contest, Daffodil International University 2014
Champion, ICT Olympiad – CSE Fest, Rajshahi University of Engineering and Technology 2012

TEACHING EXPERIENCE

Teaching breadth: More than five years as an instructor of record includes machine learning, pattern recognition, algorithms, operating systems, discrete mathematics, C++ and Java programming, computer graphics, compiler design, and web development. Additional assignments at Oklahoma State University span introductory programming, databases, computer security, data structures and algorithms, operating systems, and social issues in computing across classroom and online settings. This record supports teaching in computer science, AI, NLP, data science, and technical information systems. Instruction connects conceptual reasoning, guided practice, reproducible projects, clear assessment, and actionable feedback.

Oklahoma State University Stillwater, OK
Graduate Teaching Assistant, Department of Computer Science *August 2022 – Present*

- **Introduction to Computer Security** (Fall 2022); Instructor: Dr. Sharmin Jahan.
- **Design and Implementation of Operating Systems I** (Spring 2023, Spring 2024, and Spring 2026); Instructor: Dr. Shital Joshi.
- **Data Structures and Algorithm Analysis II** (Fall 2023); Instructor: Dr. H. K. Dai.
- **Discrete Mathematics for Computer Science** (Fall 2024); Instructor: Dr. Sachin Jain.
- **Social Issues in Computing** (Spring 2025); Instructor: Dr. Rekha Bhowmik.
- **Introduction to Database Systems** (Fall 2025); Instructor: Dr. Sadiq Albuhamood.
- **Computer Science I** (Fall 2026); supported introductory programming instruction, assessment, and student learning.
- Supported campus and online courses through laboratories, discussion, grading, student support, and instructional materials; served as lead teaching assistant for an online operating systems course and administered Canvas and zyBooks.

University of Asia Pacific Dhaka, Bangladesh
Lecturer and Instructor of Record, Department of Computer Science and Engineering *October 2018 – July 2022*

- **Theory Courses:** Machine Learning; Pattern Recognition; Design and Analysis of Algorithms; Mathematics for Computer Science; and Visual and Web Programming.
- **Laboratory Courses:** Computer Graphics; Pattern Recognition; Compiler Design; Algorithms; Object-Oriented Programming II (Java); and Visual and Web Programming.
- Held full responsibility for course planning, lectures and laboratories, projects, assessment, rubrics, and feedback; supervised undergraduate projects and contributed to outcome-based curriculum development.
- Official evaluations covered 25 course sections and 647 responses, with a weighted mean of 4.12/5 versus a 4.00/5 institutional benchmark; 20 of 25 sections exceeded their benchmarks.

Uttara University Dhaka, Bangladesh
Lecturer and Instructor of Record, Department of Computer Science and Engineering *February 2017 – October 2018*

- **Courses Taught:** Design and Analysis of Algorithms; Operating System Design; Discrete Mathematics; and Programming Language and Application II (C++).
- Delivered lectures and laboratories, prepared instructional materials and examinations, evaluated student work, and supported departmental assessment and result preparation.

STUDENT RESEARCH MENTORING

Reasoning and Artificial Intelligence (rAIson) Lab Oklahoma State University
Research Mentor; Faculty Supervisor: Dr. Atriya Sen *Spring 2026*

- **Vamsi Krishna Pavuluri: Constraint-Aware Counterfactual Editing for Aspect-Based Sentiment Analysis:** Mentored research on counterfactual generation, aspect-specific sentiment intervention, semantic preservation, experimental evaluation, and manuscript preparation.
- The project resulted in the co-authored peer-reviewed paper "Constraint-Aware Counterfactual Editing for Aspect-Based Sentiment Analysis," presented at the 2nd International Workshop on Informing Machine Learning with Knowledge Engineering for Hybrid Intelligent Systems (HHAI-KEML 2026), Brussels, Belgium.

Complex Systems Lab Oklahoma State University
Research Mentor *Fall 2023*

- **Jake Williams: Multimodal Aspect-Based Sentiment Analysis:** Mentored research on multimodal representation learning, aspect-level sentiment modeling, experimental design, model evaluation, and technical communication.

University of Asia Pacific Dhaka, Bangladesh
Student Research Mentor, Department of Computer Science and Engineering *2018 – 2022*

- **Md. Sohanuzzaman Soad, Mahady Al Hady, and Sudip Ghose: Cross-Domain Adversarial Data Augmentation:** Mentored research on GAN-based data augmentation for medical and handwriting datasets, including problem formulation, method development, experimental evaluation, and manuscript preparation.
- The project resulted in the co-authored peer-reviewed paper "Cross-Domain Adversarial Augmentation: Stabilizing GANs for Medical and Handwriting Data Scarcity," presented at IEEE RAAICON 2025.

RESEARCH COLLOQUIUM PRESENTATIONS

"Multimodal Representation Learning." Joint Ph.D. Research Colloquium, Oklahoma State University and the University of Tulsa. Spring 2025

- Surveyed joint, contrastive, and transformer-based multimodal methods, including missing modalities and adaptive modality weighting.

"Literature Review on Aspect-Based Sentiment Analysis." Joint Ph.D. Research Colloquium, Departments of Computer Science, Oklahoma State University and the University of Tulsa. Spring 2023

- Reviewed major developments in aspect extraction and sentiment classification, including pretrained language models, syntactic and graph-based methods, multitask learning, and unified generative formulations for aspect-based sentiment analysis.

"ECA-GAN: Low-Sampled Classification Using Semi-Supervised GAN Embedding with an External Classifier." Joint Ph.D. Research Colloquium, Departments of Computer Science, Oklahoma State University and the University of Tulsa. Fall 2022

- Presented a semi-supervised framework combining GAN-based data augmentation, pseudo-labeling, embedding-space analysis, and an external classifier for image classification under limited labeled-data conditions.

PROFESSIONAL SERVICE

Professional Membership: Association for Computational Linguistics (ACL), Member.

Peer Review

ACL Rolling Review (ARR), Reviewer (October cycle)	2026
ACM Transactions on Internet Technology	2026
Conference on Neural Information Processing Systems (NeurIPS), Position Paper Track	2026
International AAAI Conference on Web and Social Media (ICWSM)	2026
Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)	2025
International Conference on Computational Linguistics (COLING)	2024–2025
International Conference on Machine Learning and Applications (ICMLA)	2023–2025
International Joint Conference on Neural Networks (IJCNN)	2024

Academic Outreach and Community Service

Competitive Programming Team Coach	University of Asia Pacific; RUET Inter-University Programming Contest	2019
Volunteer	National High School Programming Contest, Rajshahi	2016
Mathematics Olympiad Volunteer	Divisional Mathematical Olympiad, Faridpur	2006

FACULTY DEVELOPMENT AND TRAINING

Improving Learning and Teaching Skills (ILTS), University of Asia Pacific May 5, 2019

Roles, Responsibilities, and Ethical Principles of University Teachers, Institutional Quality Assurance Cell, Uttara University February 24, 2018

TECHNICAL EXPERTISE

Artificial Intelligence and Language Models	Natural language processing, transformers, multimodal learning, sentiment and conversation analysis, adaptive computation, mechanistic interpretability, causal and counterfactual evaluation, reliability and bias analysis, PyTorch, Hugging Face, TensorFlow, Scikit-learn, NumPy, and pandas
Computational Tools and Software	Python, C/C++, Java, SQL, object-oriented programming, algorithms and data structures, databases, MySQL, Spark, web development, operating systems, computer security foundations, Linux, Git, Docker, and AWS
Research, Teaching, and Experiential Learning	Jupyter, controlled experiments, ablation studies, error analysis, GPU profiling, Canvas, zyBooks, project design, assessment, rubrics, and outcome-based curriculum development

PROFESSIONAL REFERENCES

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